

1. Research Objective

Central Research Question: What are the drivers of online engagement, and how can content creators boost their Instagram engagement rate?

Dataset Overview

Attribute	Details
Sample Size	5,994 Instagram posts
Number of Creators	47 top content creators
Time Period	January 2022 - December 2023
Key Variables	account, username, followers at posting, post created date, post created time, type, likes, comments, link, imagetext, postscraped

2. Methodology

2.1 Data Cleaning & Preparation

Checking duplicates (0) and missing values in the followersatposting column (19 records) were imputed using per-creator median follower counts, preserving natural variation.

A creator demographic mapping table was manually coded by reviewing each of the 47 creators' Instagram profiles, capturing Gender, Race, Location, Platform Count, Topics, Audience Age Group, Audience Gender, and Audience Primary Location. This table was merged into the post-level dataset.

The dataset was compiled and cleaned collaboratively by a three-person research team. Final datasets were merged using the link as the key.

2.2 Feature Engineering

Multiple derived variables were engineered from raw caption and timestamp data:

Variable Category	Variables Created	Method
Caption Analysis	caption_length, hashtag_count, mention_count, emoji_count	Python regex parsing
Sentiment	sentiment_compound (-1 to +1)	TextBlob NLP
Content Type Flags	ct_question, ct_emotional, ct_humor, ct_lifestyle, ct_giveaway, ct_cta, ct_sponsored, ct_announcement, ct_personal_story	Keyword pattern matching
Temporal	month, day_of_week, hour, time_slot, is_weekend	Excel Datetime extraction
Creator Metrics	creator_size, platform_count, follower_growth_rate, post_per_week	Calculated from followers
Engagement Depth	engagement_rate comment_to_like_ratio	(Comments + Likes) / Followers x 100 Comments / Likes

2.3 Exploratory Data Analysis

Initial exploration was conducted via Tableau and Excel Pivot Tables to identify distribution patterns and outliers. This was followed by a Correlation Matrix to quantify relationships (See Appendix A). At first glance, these features seemed to have almost zero impact. However, we realized their true effect was being hidden (or "mediated") by the creator's account size. We moved to a Multiple Linear Regression to peel back these layers and find the real tactical influence.

2.4 Regression Modeling (See Appendix B1, B2)

Two OLS regression models were estimated. Model 1 predicted engagement rate (ER). Model 2 predicted follower growth rate (GR), enabling comparison of which variables drive immediate engagement versus long-term audience expansion. Both models included 41 predictor variables.

Follower count was log-transformed to address right skew. Reference categories for categorical variables are Album, Afternoon (12-16), Friday, Female, Asian, Large (>10M), 18-24, and Female Audience.

Regression diagnostics were conducted to assess model validity (See Appendix C). The plots show a random "cloud" of data points, which proves our linear model is a great fit. While social media always has "viral" outliers that create some noise, the bulk of the data follows a clear, predictable path. These minor variations are normal for this type of data and don't change our final strategy.

3. Key Findings Summary

The ER model achieves a strong R-squared of 47.9%, demonstrating that engagement is highly predictable and controllable through strategic content choices. While the GR model (R-squared 7.4%) confirms that raw follower growth is influenced by other factors outside the variables, like external platform momentum, our analysis reveals that specific tactical levers, such as caption length, hashtag volume, and post frequency, are essential for converting that momentum into a loyal, permanent audience. A core strength of this study is the distinction between Observed Averages (Actual) and Statistical Influence (Impact). In several instances, raw data appeared to contradict logical recommendations due to noise from outliers and bias. By utilizing Multiple Linear Regression, we effectively isolate the true tactical impact from external variables.

3.1 Creator Demographics

Creator Size: Small creators (<1M) are the engines of growth ($B = +0.016$, $p < 0.001$), while Medium creators (1M–10M) represent the "conversion peak" with a predicted ER of 14.07%.

Creator Gender: Male creators receive significantly lower engagement than female creators ($B = -0.08$, $p < 0.001$). Transgender creators significantly outperform all groups ($B = +0.06$, $p < 0.001$). In the GR model, male creators grow their follower base faster - revealing a retention vs. acquisition trade-off.

Creator Race: Black creators significantly outperform all groups vs. the Asian baseline ($B = +0.146$, $p < 0.001$), followed by Pacific Islander ($B = +0.108$, $p < 0.001$) and White creators ($B = +0.019$, $p < 0.001$). These findings reflect observed audience behavior patterns and should be interpreted descriptively, not as value judgments. The race effect partially reverses in the GR model, suggesting high-engagement communities may grow more slowly - a depth vs. breadth trade-off.

3.2 Audience Demographics

Audience Age: Creators with 25-34 audiences receive significantly lower engagement than those with 18-24 audiences ($B = -0.089$, $p < 0.001$). Audience age is not significant in the GR model.

Audience Gender: Creators with predominantly male audiences receive significantly lower engagement ($B = -0.017$, $p < 0.001$). Audience gender is not significant in the GR model.

3.3 Post Format & Timing

Post Type: Compared to albums, photo posts receive significantly lower engagement ($B = -0.021$, $p < 0.001$) and video posts perform worse ($B = -0.059$, $p < 0.001$). Videos generate proportionally more comments relative to likes, so they win on conversation depth.

Posting Time: Night posting (0-5 am) generates significantly higher engagement than the afternoon baseline ($B = +0.014$, $p < 0.001$). Wednesday posting significantly hurts engagement ($B = -0.008$, $p < 0.001$). The optimal strategy: post important content on Saturday ($B = 0.005$, $p < 0.001$) night.

3.4 Content & Caption Features

Variable	ER Coef.	GR Coef.	Finding
Caption length	N/S	-0.000001	Tiny negative on GR
Hashtag count	-0.003	+0.001	Hurts ER, helps GR
Emoji count	-0.002	N/S	Slightly hurts ER
Sentiment/tone	N/S	N/S	No significant effect
Mention count	N/S	N/S	No significant effect
Giveaway posts	+0.029	N/S	Boosts ER only
Humor posts	+0.017	-0.007	Helps ER but hurts GR
Question posts	-0.013	N/S	Hurt ER only
Emotional posts	N/S	0.004	Boosts GR only
CTA posts	-0.013	N/S	Slightly hurts ER
Sponsored posts	N/S	N/S	No significant effect
Announcement posts	N/S	N/S	No significant effect
Personal posts	N/S	N/S	No significant effect
Lifestyle posts	N/S	N/S	No significant effect

3.5 Creator Behavior

Follower growth rate is the strongest behavioral predictor in the ER model ($B = +0.354$, $p < 0.001$). Platform count also positively predicts engagement ($B = +0.007$, $p < 0.001$), indicating cross-platform presence builds more loyal core audiences. Increasing posting frequency per week generates higher engagement rate ($B = +0.003$, $p < 0.001$).

3.6 The Engagement vs. Growth Trade-off

One of the most original findings is the systematic tension between variables that drive engagement versus follower growth, showcasing throughout the variables above. Running two separate models on the same dataset revealed that these two outcomes require different strategies.

Core Insight: Engagement and Growth are not independent; they are a feedback loop. While identity sets the starting point, content decisions act as the accelerator.

4. Strategic Recommendations

Creator Size	Primary Metric	Key Strategies
Small (<1M)	Follower Growth Rate	• Prioritize short, high-impact captions. • Increase emotional storytelling while reducing reliance on humor • Utilize hashtags intentionally to drive discoverability • Focus resources on maximizing Instagram presence • Adopt video as the lead content type to deepen conversation
Medium (1M-10M)	Engagement Rate	• Lead with humor and giveaways; reduce CTAs and questions • Utilize album/carousel posts as a high-engagement primary type • Maintain high-frequency weekly posting • Minimize hashtag density to reduce visual clutter • Maintain a consistent cross-platform brand presence • Minimize emoji usage for a cleaner caption aesthetic • Prioritize Saturday uploads over low-efficiency Wednesdays
Large (>10M)	Comment-to-Like Ratio	Leverage video to drive meaningful dialogue over passive likes
Everyone		• Target high-attention windows: 17:00–20:00 or 00:00–05:00

For Large creators, we propose the comment-to-like ratio as a more meaningful success metric. Our data shows Large creators have both the lowest engagement rate (baseline) and the lowest comment-to-like ratio (0.004 vs 0.024 for Medium), suggesting their audiences are the most passive.

5. Limitations & Future Research

5.1 Analytical Limitations

- OLS assumption violations: Residual diagnostics revealed heteroscedasticity and right-skewed errors driven by viral outlier posts. Future research could apply log-transformed engagement rates or robust standard errors to address these.
- Multicollinearity: High correlation between log_followers and creator_size required removing one variable. The condition number (1.17e+16) indicates remaining collinearity concerns.
- Cross-sectional limitations: The 2022-2023 period may not generalize to future years, given Instagram algorithm changes.
- Clustering: Posts nested within creators (avg. 127 posts per creator) violates independence. A multilevel regression model would provide more precise estimates.

- Manual demographic coding: Creator demographics were coded by the research team based on observable profile characteristics - perceived categories, not self-reported.

5.2 Future Research

- Fixed effects regression to control for creator-level unobserved heterogeneity
- Hierarchical multilevel model separating within-creator from between-creator variation
- Large creator comment-to-like regression to formally test deepened engagement strategies
- Longitudinal analysis tracking creators across 3-5 years for causal claims
- Platform comparison extending to TikTok, YouTube, and X for the same creators

6. Conclusion

This analysis establishes that Instagram success is not the result of random virality, but a measurable outcome of aligning Creator Identity with Strategic Content Execution. By utilizing a dual-model framework, we have demonstrated that Engagement Rate and Follower Growth are distinct objectives that require different strategic focuses.

The overarching data trend reveals that Creator Demographics and Account Size serve as the "performance floor," establishing the baseline potential for reach and interaction. However, Content Strategy serves as the "performance ceiling," the essential lever that allows a creator to outperform their peers, stabilize their feed, and bypass the noise of the platform's algorithm.

Our findings lead to three macro-level conclusions for the industry:

- Strategic alignment over guesswork: Success is found when creators match their specific phase of the Creator Lifecycle (Reach vs. Depth) to the appropriate KPIs, ensuring optimized ROI for both creators and brand partners.
- Quality-driven stability: The data favors a transition from shallow growth strategies to community-deepening tactics. Building a foundation of high-value interaction is the only sustainable way.
- The multiplier effect: While identity is a strong predictor of engagement, data-backed optimization acts as a force multiplier. Strategic execution allows creators to turn raw platform momentum into a loyal, high-converting audience.

Ultimately, the data tells a story of intentionality. In an increasingly crowded digital landscape, the creators who win are those who move beyond surface-level analytics to understand the deeper statistical drivers of their influence. This framework provides the roadmap to turn that data into a long-term competitive advantage.

Statement of AI Assistance: Generative AI (Claude, Gemini, and ChatGPT) was utilized as a technical co-pilot throughout this project. Its primary roles included optimizing Python scripts for efficiency, clarifying complex statistical concepts, and refining the report's professional tone. While AI assisted in processing and phrasing, all core hypotheses, data interpretations, and final strategic recommendations were developed and verified by the human team to ensure accuracy and contextual relevance.

7. Appendix

Appendix A: The Correlation Matrix (Heatmap)

Appendix A confirms the absence of significant multicollinearity, ensuring that each tactical driver (e.g., Emojis, Albums, Humor) exerts an independent influence on the results.



Appendix B: Regression Models

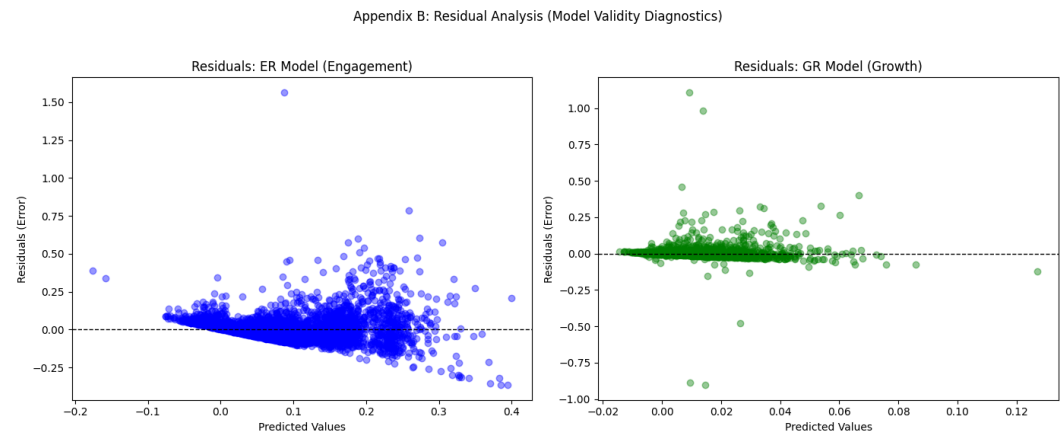
Appendix B1: Engagement Rate (ER)

OLS Regression Results						
Dep. Variable:	engagement_rate	R-squared:	0.479			
Model:	OLS	Adj. R-squared:	0.476			
Method:	Least Squares	F-statistic:	140.1			
Date:	Tue, 31 Mar 2026	Prob (F-statistic):	0.00			
Time:	06:24:23	Log-Likelihood:	7186.1			
No. Observations:	5974	AIC:	-1.429e+04			
Df Residuals:	5934	BIC:	-1.402e+04			
Df Model:	39					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.1264	0.007	18.623	0.000	0.113	0.140
C(type)[T.Photo]	-0.0209	0.002	-8.714	0.000	-0.026	-0.016
C(type)[T.Video]	-0.0591	0.006	-9.965	0.000	-0.071	-0.048
C(time_slot)[T.Evening (17-20)]	0.0059	0.003	1.920	0.055	-0.000	0.012
C(time_slot)[T.Late Night (21-23)]	0.0036	0.003	1.048	0.295	-0.003	0.010
C(time_slot)[T.Morning (6-11)]	-0.0011	0.004	-0.282	0.778	-0.009	0.006
C(time_slot)[T.Night (0-5)]	0.0139	0.004	3.810	0.000	0.007	0.021
C(day_of_week)[T.Monday]	0.0019	0.003	0.549	0.583	-0.005	0.009
C(day_of_week)[T.Saturday]	0.0052	0.002	2.431	0.015	0.001	0.009
C(day_of_week)[T.Sunday]	-0.0015	0.002	-0.730	0.465	-0.006	0.003
C(day_of_week)[T.Thursday]	-0.0026	0.003	-0.756	0.450	-0.009	0.004
C(day_of_week)[T.Tuesday]	-0.0054	0.003	-1.540	0.124	-0.012	0.001
C(day_of_week)[T.Wednesday]	-0.0001	0.004	-2.308	0.021	-0.015	-0.001
C(is_weekend)[T.Weekend]	0.0037	0.002	1.792	0.073	-0.000	0.008
C(creator_size)[T.Medium (1M-10M)]	0.0143	0.005	2.633	0.008	0.004	0.025
C(creator_size)[T.Small (<1M)]	-0.0502	0.010	-5.118	0.000	-0.069	-0.031
C(creator_gender)[T.Male]	-0.0808	0.003	-24.363	0.000	-0.087	-0.074
C(creator_gender)[T.Transgender]	0.0599	0.007	8.327	0.000	0.046	0.074
C(creator_race)[T.Black]	0.1463	0.006	26.582	0.000	0.136	0.157
C(creator_race)[T.Pacific Islander]	0.1085	0.007	15.064	0.000	0.094	0.123
C(creator_race)[T.White]	0.0194	0.005	4.178	0.000	0.010	0.029
C(audience_age)[T.25-34]	-0.0094	0.004	-24.154	0.000	-0.097	-0.082
C(audience_gender)[T.Male]	-0.0176	0.003	-5.034	0.000	-0.024	-0.011
follower_growth_rate_centered	0.3538	0.027	13.028	0.000	0.301	0.407
log_followers_centered	-0.0350	0.002	-16.478	0.000	-0.039	-0.031
caption_length_centered	1.156e-05	6.04e-06	1.913	0.056	-2.86e-07	2.34e-05
hashtag_count_centered	-0.0033	0.001	-5.222	0.000	-0.004	-0.002
mention_count_centered	-9.882e-05	0.001	-0.185	0.853	-0.001	0.001
emoji_count_centered	-0.0028	0.001	-4.837	0.000	-0.004	-0.002
sentiment_compound2_centered	-0.0036	0.003	-1.336	0.181	-0.009	0.002
post_per_week_centered	0.0033	0.000	8.947	0.000	0.003	0.004
platform_count_centered	0.0070	0.001	5.932	0.000	0.005	0.009
ct_cta	-0.0132	0.004	-2.957	0.003	-0.022	-0.004
ct_giveaway	0.0292	0.008	3.688	0.000	0.014	0.045
ct_sponsored	0.0072	0.005	1.547	0.122	-0.002	0.016
ct_question	-0.0126	0.003	-3.889	0.000	-0.019	-0.006
ct_announcement	0.0012	0.008	0.152	0.879	-0.014	0.016
ct_personal_story	0.0023	0.006	0.377	0.706	-0.010	0.014
ct_lifestyle	0.0065	0.004	1.752	0.080	-0.001	0.014
ct_humor	0.0171	0.005	3.342	0.001	0.007	0.027
ct_emotional	0.0029	0.003	0.931	0.352	-0.003	0.009

Appendix B2: Growth Rate (GR)

OLS Regression Results						
Dep. Variable:	follower_growth_rate	R-squared:	0.074			
Method:	Least Squares	Adj. R-squared:	0.068			
Date:	Tue, 31 Mar 2026	F-statistic:	12.16			
Time:	06:24:23	Prob (F-statistic):	7.49e-73			
No. Observations:	5974	Log-Likelihood:	11680.1			
Df Residuals:	5934	AIC:	-2.328e+04			
Df Model:	39	BIC:	-2.301e+04			
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.0034	0.003	1.056	0.291	-0.003	0.010
C(type)[T.Photo]	6.946e-05	0.001	0.061	0.951	-0.002	0.002
C(type)[T.Video]	0.0061	0.003	2.168	0.030	0.001	0.012
C(time_slot)[T.Evening (17-20)]	0.0011	0.001	0.747	0.455	-0.002	0.004
C(time_slot)[T.Late Night (21-23)]	0.0006	0.002	0.343	0.731	-0.003	0.004
C(time_slot)[T.Morning (6-11)]	0.0006	0.002	0.328	0.743	-0.003	0.004
C(time_slot)[T.Night (0-5)]	0.0034	0.002	1.966	0.049	9.38e-06	0.007
C(day_of_week)[T.Monday]	0.0020	0.002	1.216	0.224	-0.001	0.005
C(day_of_week)[T.Saturday]	1.803e-05	0.001	0.018	0.986	-0.002	0.002
C(day_of_week)[T.Sunday]	0.0002	0.001	0.238	0.812	-0.002	0.002
C(day_of_week)[T.Thursday]	-0.0001	0.002	-0.063	0.950	-0.003	0.003
C(day_of_week)[T.Tuesday]	0.0015	0.002	0.894	0.371	-0.002	0.005
C(day_of_week)[T.Wednesday]	-0.0012	0.002	-0.698	0.485	-0.004	0.002
C(is_weekend)[T.Weekend]	0.0003	0.001	0.262	0.794	-0.002	0.002
C(creator_size)[T.Medium (1M-10M)]	0.0059	0.003	2.298	0.022	0.001	0.011
C(creator_size)[T.Small (<1M)]	0.0133	0.005	2.868	0.004	0.004	0.022
C(creator_gender)[T.Male]	0.0094	0.002	5.779	0.000	0.006	0.013
C(creator_gender)[T.Transgender]	0.0098	0.003	2.868	0.004	0.003	0.016
C(creator_race)[T.Black]	-0.0035	0.003	-1.270	0.204	-0.009	0.002
C(creator_race)[T.Pacific Islander]	-0.0114	0.003	-3.288	0.001	-0.018	-0.005
C(creator_race)[T.White]	-0.0064	0.002	-2.941	0.003	-0.011	-0.002
C(audience_age)[T.25-34]	-0.0022	0.002	-1.187	0.235	-0.006	0.001
C(audience_gender)[T.Male]	-0.0032	0.002	-1.946	0.052	-0.006	2.41e-05
engagement_rate_centered	0.0706	0.006	13.028	0.000	0.057	0.090
log_followers_centered	0.0025	0.001	2.411	0.016	0.000	0.004
caption_length_centered	-6.846e-06	2.85e-06	-2.405	0.016	-1.24e-05	-1.26e-06
hashtag_count_centered	0.0013	0.000	4.321	0.000	0.001	0.002
mention_count_centered	0.0001	0.000	0.524	0.600	-0.000	0.001
emoji_count_centered	0.0003	0.000	1.010	0.312	-0.000	0.001
sentiment_compound2_centered	-0.0002	0.001	-0.176	0.860	-0.003	0.002
post_per_week_centered	0.0002	0.000	1.240	0.215	-0.000	0.001
platform_count_centered	-0.0014	0.001	-2.505	0.012	-0.002	-0.000
ct_cta	0.0025	0.002	1.173	0.241	-0.002	0.007
ct_giveaway	-0.0017	0.004	-0.461	0.645	-0.009	0.006
ct_sponsored	-5.931e-05	0.002	-0.027	0.978	-0.004	0.004
ct_question	0.0022	0.002	1.473	0.141	-0.001	0.005
ct_announcement	0.0020	0.004	0.542	0.588	-0.005	0.009
ct_personal_story	0.0013	0.003	0.444	0.657	-0.004	0.007
ct_lifestyle	0.0003	0.002	0.144	0.886	-0.003	0.004
ct_humor	-0.0067	0.002	-2.761	0.006	-0.011	-0.002
ct_emotional	0.0038	0.001	2.539	0.011	0.001	0.007

Appendix C: Residual Analysis Plots



Appendix B displays the random residual distribution, confirming that our linear model specifications are appropriate and that the findings are not biased by non-linear patterns.

Appendix D: Feature Engineering

Variable	Formula	Notes
A. Engagement Metrics		
engagement rate	$(likes + comments) / followersatposting \times 100$	Main outcome variable
comment to like ratio	$comments / (likes + 1)$	#+1 prevents division by zero
log followers	$np.log(followersatposting + 1)$	Log transform to address right skew
B. Follower Growth		
follower growth pct	<code>pct_change()</code> grouped by username	Percentage change from previous post per creator
follower growth rate	$(Max\ followers - Min\ followers) / Min\ followers \times 100$	Total growth across dataset period per creator
C. Caption Text Features		
caption length	<code>len()</code> on <code>postscraped</code>	Character count including spaces
hashtag count	<code>regex r'#\w+'</code>	Count of all # occurrences
mention count	<code>regex r'@\w+'</code>	Count of all @ mentions
emoji count	<code>emoji.EMOJI_DATA</code> lookup	Count characters in emoji dictionary
sentiment compound	<code>TextBlob .sentiment.polarity</code>	-1 (negative) to +1 (positive)
D. Temporal Variables		
month	<code>datetime</code> extraction	January through December
day of week	<code>datetime</code> extraction	Monday through Sunday
hour	<code>datetime</code> extraction	0 through 23
time slot	binned from hour	Morning (6-11), Afternoon (12-16), Evening (17-20), Late Night (21-23), Night (0-5)
is weekend	<code>day of week</code> in Saturday/Sunday	0 or 1
E. Creator Size & Tier		
influencer tier	binned from <code>followersatposting</code>	Nano (<10K), Micro (10K-100K), Mid-Tier (100K-500K), Macro (500K-1M), Mega (>1M)
creator size	binned from <code>followersatposting</code>	Small (<1M), Medium (1M-10M), Large (>10M)
platform count	<code>len(split(',')</code> on Social Media Platforms	Count of comma-separated platforms
G. Content Type Flags — Binary ct_variables (from <code>postscraped</code>)		
ct_giveaway	<code>giveaway, give away, win, winner, contest, prize, enter to win, raffle</code>	
ct_sponsored	<code>#ad, #sponsored, #paid, #partner, paid partnership, in collaboration with</code>	
ct_qta	<code>link in bio, swipe, comment below, tag a friend, follow me, share this, check out, go watch, make sure</code>	
ct_question	<code>regex r'?' — any question mark</code>	
ct_announcement	<code>excited to announce, announcing, just launched, now available, out now, streaming now, new episode, just dropped, dropping</code>	
ct_personal_story	<code>throwback, throw back, tbt, remember when, back in the day, birthday, bday, baby picture, high school, growing up, i used to</code>	
ct_lifestyle	<code>oord, outfit, travel, vacation, trip to, beach, pool, summer, winter, holiday, christmas, halloween, coachella, festival</code>	
ct_humor	<code>laughing emoji, rolling on floor emoji, lol, lmao, roast, funny, comedy, joke, prank, meme</code>	
ct_emotional	<code>dream come true, grateful, blessed, thankful, love you, i love, miss you, crying, heart emoji, pleading emoji, so proud, proud of</code>	
H. Creator Demographics — Manually Input		
creator_gender	Manual profile review	Female, Male, Transgender
creator_race	Manual profile review	Asian, Black, White, Pacific Islander
creator_location	Manual profile review	Country/region
creator_content_type	Manual categorization	Gaming & Streaming, Fashion & Beauty, Comedy & Entertainment, Lifestyle, Fitness & Sport, Food & Experiment, Music & Arts, Family & Parenting, Business & Finance, Adult & Niche Content
I. Audience Demographics — From Influencer Profile Sheet		
audience_age	Influencer profile Excel — Age Group column	
audience_gender	Influencer profile Excel — Gender column	
audience_location	Influencer profile Excel — Primary Location column	